

Internet Security Flow Analysis

Understanding Internet Communication Through a Cybersecurity Perspective

Technical Project Report

Prepared by

MD Siam Babon

Aspiring Cybersecurity Analyst

Areas of Focus

- Security Operations (SOC)
- Microsoft Azure
- Cloud Security
- Networking Fundamentals
- Windows Security
- Linux Fundamentals

Project Version

Version 2.0

July 2026

Project Repository

GitHub

<https://github.com/Siambabon/internet-security-analysis>

Professional Statement

"Cybersecurity begins with understanding how systems communicate, where trust exists, and how defensive controls protect modern digital environments."

© 2026 MD Siam Babon

This project was developed for educational and professional portfolio purposes.

</div>

PAGE 2 — TABLE OF CONTENTS

Table of Contents

Section	Page
Cover Page	1
Table of Contents	2
Introduction & Project Background	3
Project Objectives & Scope	4
Understanding Internet Communication	5
Internet Communication Workflow	6
Project Architecture Analysis	7
Attack Surface Analysis	8
Defensive Security Controls	9
Applying the CIA Triad	10
Real-World Security Scenario	11
Skills Demonstrated	12
Learning Reflection	13

References	14
Conclusion & About the Author	15

PAGE 3 — INTRODUCTION & PROJECT BACKGROUND

1. Introduction

Modern society depends on the Internet for communication, banking, education, healthcare, cloud computing, and countless other digital services. Every interaction with a website or online application involves multiple interconnected systems working together behind the scenes. Although these processes appear simple to users, they rely on a series of trust relationships that must be protected against cyber threats.

Understanding these communication processes is one of the most important foundations of cybersecurity. Before security professionals can detect attacks, investigate incidents, or design secure systems, they must first understand how information travels across networks and how different technologies interact throughout the communication process.

This project was developed to build that foundational understanding by examining the complete journey of an internet request—from the moment a user enters a website address to the point where the requested information is returned to the browser. Instead of studying networking and cybersecurity as separate disciplines, this project combines both perspectives into a single workflow that highlights where communication occurs, where trust exists, and where security risks may emerge.

Throughout the project, each stage of internet communication is analyzed from a defensive security perspective. Common attack surfaces are identified, practical security controls are introduced, and real-world examples demonstrate how attackers attempt to exploit weaknesses within modern systems. By connecting networking concepts with cybersecurity principles, this project encourages analytical thinking and promotes a stronger understanding of how defensive strategies help protect digital environments.

This report is intended as a foundational cybersecurity project that demonstrates practical learning, structured analysis, and continuous professional development. It serves as part of a growing technical portfolio focused on Security Operations (SOC), Microsoft Azure, Cloud Security, and modern defensive cybersecurity practices.

PAGE 4 — PROJECT OBJECTIVES & SCOPE

2. Project Objectives & Scope

2.1 Project Purpose

The primary purpose of this project is to develop a practical understanding of how internet communication functions from a cybersecurity perspective. Rather than focusing solely on networking concepts or individual security technologies, this project examines the complete communication process to understand how trust is established between systems, where vulnerabilities may arise, and how defensive controls help reduce security risks.

This project was designed as a learning-oriented technical analysis that bridges the gap between networking fundamentals and modern defensive cybersecurity practices. It emphasizes analytical thinking, security awareness, and the practical application of foundational cybersecurity concepts.

2.2 Project Objectives

The objectives of this project are to:

- Develop a structured understanding of the lifecycle of internet communication.
 - Identify common attack surfaces throughout the communication process.
 - Analyze how attackers may exploit weaknesses at different stages.
 - Examine the role of defensive security controls in protecting digital systems.
 - Apply fundamental cybersecurity principles to real-world communication scenarios.
 - Strengthen analytical thinking from both an attacker's and a defender's perspective.
 - Build a strong technical foundation for future learning in Security Operations (SOC), Cloud Security, and Microsoft Azure.
-

2.3 Project Scope

This project focuses on the defensive analysis of internet communication and does **not** involve offensive security techniques or unauthorized system access.

The scope includes:

- Internet communication workflow

- DNS resolution
- Client–server communication
- Common cyber threats
- Defensive security controls
- Security principles
- Risk awareness
- Foundational cybersecurity concepts

The project is intended for educational purposes and serves as part of a professional cybersecurity portfolio.

PAGE 5 — UNDERSTANDING INTERNET COMMUNICATION

3. Understanding Internet Communication

Every interaction on the Internet begins with a simple action performed by a user. Whether accessing a website, checking email, or using a cloud application, the request travels through multiple interconnected systems before reaching its destination. Each component within this process performs a specific function, and together they enable secure and reliable communication across the Internet.

Although this communication often appears instantaneous, it involves several technologies working together behind the scenes. Domain Name System (DNS) services translate human-readable domain names into IP addresses, browsers generate requests using standard communication protocols, routers forward packets across networks, firewalls inspect and filter traffic, and web servers process incoming requests before returning the appropriate response.

From a cybersecurity perspective, every stage introduces a trust relationship between systems. Whenever one system trusts another, there is an opportunity for attackers to exploit that trust through techniques such as spoofing, interception, unauthorized access, or malicious input. Understanding these trust relationships is essential because cybersecurity is not only about preventing attacks—it is also about understanding how systems communicate so that security controls can be applied effectively.

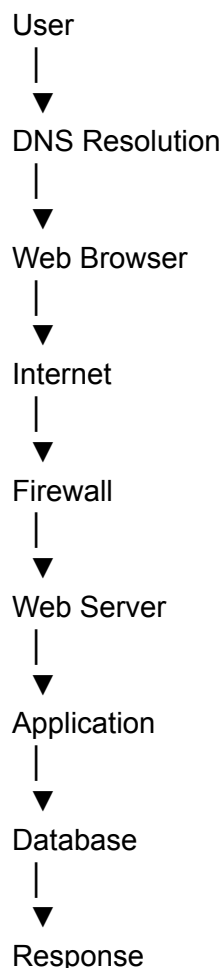
A strong understanding of internet communication provides the foundation required for Security Operations Centers (SOC), incident response, cloud security, vulnerability management, and many other cybersecurity disciplines. For this reason, networking

fundamentals remain one of the most important building blocks for every cybersecurity professional.

PAGE 6 — INTERNET COMMUNICATION WORKFLOW

4. Internet Communication Workflow

The following workflow illustrates the typical path followed when a user accesses a website over the Internet.



Each stage within this workflow performs a unique function while introducing different security considerations.

User

The communication process begins when a user enters a website address or submits a request through a web browser. User behavior plays a significant role in cybersecurity because human error remains one of the most common causes of security incidents.

DNS Resolution

The Domain Name System translates the requested domain name into an IP address so that the browser can locate the correct server. Since DNS establishes trust between the user and the destination, it is frequently targeted by attackers through techniques such as DNS spoofing and cache poisoning.

Web Browser

The browser creates an HTTP or HTTPS request and initiates communication with the destination server. Modern browsers also perform certificate validation and enforce various security policies that help protect users during web communication.

Internet

The request travels through multiple routers, switches, and service providers before reaching its destination. During this stage, data may become vulnerable if secure communication protocols are not properly implemented.

Firewall

Firewalls inspect incoming and outgoing traffic according to predefined security rules. Their primary purpose is to reduce unauthorized access while allowing legitimate communication to continue.

Web Server

The server receives the client's request, processes it, and forwards the necessary information to the application layer. Secure server configuration is critical for reducing the risk of unauthorized access and system compromise.

Application & Database

The application processes business logic and interacts with the database to retrieve or store information. Secure coding practices and input validation are essential to prevent attacks such as SQL Injection and unauthorized data access.

Response

Finally, the server returns the requested information to the user. Throughout this process, encryption, authentication, and integrity verification help ensure that the response remains secure and trustworthy during transmission.

PAGE 7 — PROJECT ARCHITECTURE ANALYSIS

5. Project Architecture Analysis

Understanding the Communication Flow

The architecture presented in this project illustrates the journey of a typical web request and highlights how multiple technologies interact to deliver information securely between a user and a web application. Although this process usually completes within milliseconds, every stage introduces trust relationships, communication protocols, and potential security risks that must be carefully managed.

<div align="center">

[Insert Project Architecture Diagram Here]

Figure 1. Internet communication workflow illustrating the relationship between the user, networking components, security controls, application services, and data storage.

</div>

The communication process begins when a user requests access to a website by entering a domain name into a web browser. The browser relies on the Domain Name System (DNS) to resolve the domain into an IP address, allowing the request to locate the correct destination server.

After DNS resolution, the browser generates an HTTP or HTTPS request and sends it across multiple network devices before reaching the destination environment. During transmission, routers, switches, and Internet Service Providers forward packets while firewalls inspect traffic according to predefined security policies.

Once the request reaches the web server, it is processed by the application layer. If required, the application retrieves or stores information within a database before generating a response that travels back through the same communication path to the user's browser.

Although the workflow appears linear, every component depends on trust. A compromised DNS server, an exposed web application, or an improperly configured firewall can significantly affect the confidentiality, integrity, and availability of information throughout the communication process.

From a cybersecurity perspective, understanding these relationships is essential because effective security depends on protecting every stage of communication rather than focusing on a single technology.

PAGE 8 — ATTACK SURFACE ANALYSIS

6. Attack Surface Analysis

Understanding Potential Threats

Every connected system exposes some level of attack surface. An attack surface represents any point where an attacker may attempt to interact with, exploit, or compromise a system. The objective of defensive cybersecurity is not to eliminate every possible risk—an impossible task—but to reduce the attack surface through secure design, proper configuration, continuous monitoring, and layered security controls.

The following table summarizes common attack opportunities identified throughout the communication workflow.

Communication Stage	Potential Threat	Security Impact	Risk Level
DNS Resolution	DNS Spoofing	Redirects users to malicious destinations	High
Network Communication	Man-in-the-Middle (MITM)	Data interception or modification	High
Firewall Configuration	Misconfigured Rules	Unauthorized network access	Medium
Web Server	Service Misconfiguration	System compromise	High
Application Layer	Injection Attacks	Data manipulation and unauthorized actions	High
Database	Unauthorized Access	Exposure of sensitive information	High
User	Phishing & Social Engineering	Credential theft	High

Security Perspective

One important lesson from this analysis is that attackers rarely target only one technology. Instead, they attempt to identify the weakest point within the communication process. In many cases, human behavior, insecure configurations, or outdated software provide easier opportunities than sophisticated technical exploits.

For this reason, modern cybersecurity emphasizes **defense in depth**, where multiple independent security controls work together to reduce the likelihood and impact of successful attacks.

Attack surface analysis also supports risk-based decision-making by helping organizations prioritize security improvements according to business impact, likelihood of exploitation, and available defensive capabilities.

PAGE 9 — DEFENSIVE SECURITY CONTROLS

7. Defensive Security Controls

Cybersecurity is fundamentally concerned with reducing risk rather than eliminating it completely. Organizations implement multiple defensive controls to protect systems, detect suspicious activity, and maintain the confidentiality, integrity, and availability of information.

The following controls play a critical role throughout the internet communication workflow.

Security Control	Primary Purpose
Firewall	Filters and controls network traffic according to security policies.
HTTPS / TLS	Encrypts communication between clients and servers to protect data in transit.
Multi-Factor Authentication (MFA)	Adds an additional layer of identity verification beyond passwords.
Secure Configuration	Reduces unnecessary exposure by disabling insecure services and applying security best practices.
Input Validation	Prevents malicious user input from affecting application behavior.

Monitoring & Logging	Records system activity and supports incident detection and forensic investigations.
Virtual Private Network (VPN)	Protects remote communication through encrypted network tunnels.

Defense in Depth

Modern organizations rarely depend on a single security solution. Instead, they implement multiple layers of protection so that if one control fails, additional controls continue protecting the environment.

For example, even if a phishing attack successfully steals a user's password, Multi-Factor Authentication (MFA) may prevent unauthorized account access. Likewise, encrypted HTTPS communication helps protect sensitive information even when network traffic passes through untrusted infrastructure.

This layered security approach significantly improves organizational resilience and forms one of the fundamental principles of modern cybersecurity architecture.